Type MS · Reading a sine model: values, biggest and smallest

Answer 1 of 4 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b.ii in this mark scheme extract

Number		
1 (a)	$t = \frac{10}{3}$ $P = 3 + 2\sin\frac{5\pi}{4}$ $= 3 - \sqrt{2}$ oe (e.g. $3 - \frac{2}{\sqrt{2}}$)	M1 A1 (2)
(b) (i)	5	B1
(ii)	1	B1 (2)
(c)	$4 = 3 + 2\sin\left(\frac{3\pi t}{8}\right)$ $\frac{1}{2} = \sin\left(\frac{3\pi t}{8}\right)$ $\frac{\pi}{6} = \left(\frac{3\pi t}{8}\right)$ $t = \frac{4}{9}$ oe	M1 M1 A1 (3)
		[7]

Part	Mark	Additional Guidance
(a)	M1	Correct substitution of $t = \frac{10}{3}$, leading to a value for P, simplification not required.
	A1	Answer stated, oe exact value.
(c)	M1	Correctly substitutes the value of $P = 4$ and rearranges to give an expression of the form $a = \sin\left(\frac{3\pi t}{8}\right)$ **Do not allow this mark if $a > 1$ or $a < -1$**
	M1	Correctly uses the inverse sin function to arrive at $b = \left(\frac{3\pi t}{8}\right)$ and solves to find a value of $\frac{3\pi t}{8}$, allow this value to be in degrees If the inverse sin function is not shown, then the value of the angle obtained must be correct for their *b*.
	A1	oe

Type MS · Reading a sine model: values, biggest and smallest

Answer 1 of 4 · Question 1 part b.ii · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Reading a sine model: values, biggest and smallest

Answer 2 of 4 · Question 1 part a · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

1 (a)	$t = \frac{10}{3}$ $P = 3 + 2\sin\frac{5\pi}{4}$ $= 3 - \sqrt{2}$ oe (e.g. $3 - \frac{2}{\sqrt{2}}$)	M1 A1 (2)
(b) (i)	5	B1
(ii)	1	B1 (2)
(c)	$4 = 3 + 2\sin\left(\frac{3\pi t}{8}\right)$ $\frac{1}{2} = \sin\left(\frac{3\pi t}{8}\right)$ $\frac{\pi}{6} = \left(\frac{3\pi t}{8}\right)$ $t = \frac{4}{9}$ oe	M1 M1 A1 (3)
		[7]

Part	Mark	Additional Guidance
(a)	M1	Correct substitution of $t = \frac{10}{3}$, leading to a value for P, simplification not required.
	A1	Answer stated, oe exact value.
(c)	M1	Correctly substitutes the value of $P = 4$ and rearranges to give an expression of the form $a = \sin\left(\frac{3\pi t}{8}\right)$ **Do not allow this mark if $a > 1$ or $a < -1$**
	M1	Correctly uses the inverse sin function to arrive at $b = \left(\frac{3\pi t}{8}\right)$ and solves to find a value of $\frac{3\pi t}{8}$, allow this value to be in degrees If the inverse sin function is not shown, then the value of the angle obtained must be correct for their b.
	A1	oe

Type MS · Reading a sine model: values, biggest and smallest

Answer 2 of 4 · Question 1 part a · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Reading a sine model: values, biggest and smallest

Answer 3 of 4 · Question 1 part b.i · 0 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

1 (a)	$t = \frac{10}{3}$ $P = 3 + 2\sin\frac{5\pi}{4}$ $= 3 - \sqrt{2}$ oe (e.g. $3 - \frac{2}{\sqrt{2}}$)	M1 A1 (2)
(b) (i)	5	B1
(ii)	1	B1 (2)
(c)	$4 = 3 + 2\sin\left(\frac{3\pi t}{8}\right)$ $\frac{1}{2} = \sin\left(\frac{3\pi t}{8}\right)$ $\frac{\pi}{6} = \left(\frac{3\pi t}{8}\right)$ $t = \frac{4}{9}$ oe	M1 M1 A1 (3)
		[7]

Part	Mark	Additional Guidance
(a)	M1	Correct substitution of $t = \frac{10}{3}$, leading to a value for P, simplification not required.
	A1	Answer stated, oe exact value.
(c)	M1	Correctly substitutes the value of $P = 4$ and rearranges to give an expression of the form $a = \sin\left(\frac{3\pi t}{8}\right)$ **Do not allow this mark if $a > 1$ or $a < -1$**
	M1	Correctly uses the inverse sin function to arrive at $b = \left(\frac{3\pi t}{8}\right)$ and solves to find a value of $\frac{3\pi t}{8}$, allow this value to be in degrees If the inverse sin function is not shown, then the value of the angle obtained must be correct for their b.
	A1	oe

Type MS · Reading a sine model: values, biggest and smallest

Answer 3 of 4 · Question 1 part b.i · 0 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Reading a sine model: values, biggest and smallest

Answer 4 of 4 · Question 12 part a · 1 mark

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
12 (a)	$(2\cos x = 0)$ $\quad (x =)\dfrac{\pi}{2}$ or $90°$	B1 (1)
(b)	$(2\cos x = 2\sin x)$ $\quad \tan x = 1$	M1
	$x = \dfrac{\pi}{4}$ or $45°$	A1 (2)
(c)	$\int_{(0)}^{("\frac{\pi}{4}")} (2\sin x)dx + \int_{("\frac{\pi}{4}")}^{("\frac{\pi}{2}")} (2\cos x)dx$	M1
	$\left[-2\cos x\right]_0^{\frac{\pi}{4}} + \left[2\sin x\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}}$	A1
	$\left(-\sqrt{2}+2\right)+\left(2-\sqrt{2}\right) = 4-2\sqrt{2}$	dM1
	$= 4-\sqrt{8}$ cao	A1cao cso (4) [7]

Part	Mark	Additional Guidance
(a)	B1	For $\dfrac{\pi}{2}$ or 90 degrees. Can also be shown as a coordinate – ignore any incorrect y coordinate.
(b)	M1	For $\tan x = 1$
	A1	For $\dfrac{\pi}{4}$ or 45 degrees. Can also be shown as a coordinate – ignore any incorrect y coordinate.
(c)	M1	For both integrals correctly shown, with an addition sign between them. Limits need not be shown. Must be 2 integrals shown. Can't be shown as one integral with incorrect limits.
	A1	For both functions correctly integrated. Limits need not be shown.
	dM1	For their limits clearly and correctly substituted in or for the numerical expression(s) shown in the MS. If mark awarded for substitution, both integrated expressions must have both limits correctly substituted. 0 must be the lower limit on the first integral. Allow ft of their $\dfrac{\pi}{4}$ (must be the upper limit on the first integral and the lower limit on the second and can be in degrees) and their $\dfrac{\pi}{2}$ (must be the upper limit on the second integral and can be in degrees).
	A1	cao cso A0 if degrees used in part c
		Note, can also be completed as either integral doubled – symmetry. M1 – correct integral stated with multiply by 2 evident or implicit later. A1 correctly integrated dM1 – as for main scheme, the multiply by 2 must be clearly shown. A1 as main scheme

Type MS · Reading a sine model: values, biggest and smallest

Answer 4 of 4 · Question 12 part a · 1 mark

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract